Chapter II.2, Week 2

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2.1. Let A be a ring, $X = \operatorname{Spec} A$ its spectrum. Fix an element $f \in A$, $D(f) = V(f)^c \subset X$ the associated distinguished open set. Show that $(D(f), \mathcal{O}_X|_{D(f)}) \cong \operatorname{Spec} A_f$ as locally ringed spaces.

Proof. Let $Y = \operatorname{Spec} A_f$, $S = \{f^n\}_{n\geqslant 1}$, and $\phi : A \to S^{-1}A = A_f$ be the natural map $a \mapsto a/1$. Note that $f^n \in \mathfrak{p}$ prime for some $n \geqslant 1$ iff $f \in \mathfrak{p}$—that is, $S \cap \mathfrak{p} = \emptyset$ iff $\mathfrak{p} \notin D(f)$. Thus, AM 3.11.iv gives us a bijection

$$\begin{aligned} D(f) &\longleftrightarrow Y \\ \mathfrak{p} &\longrightarrow S^{-1}\mathfrak{p}, \\ \phi^*(\mathfrak{q}) &\longleftarrow \mathfrak{q}. \end{aligned}$$

Clearly the map $\mathfrak{q} \mapsto \phi^*(\mathfrak{q})$ is continuous, so it suffices to show that $\mathfrak{p} \mapsto S^{-1}\mathfrak{p}$ is continuous.

Let $V(\mathfrak{b}) \subset A_f$ be closed, and denote $\psi : \mathfrak{p} \mapsto S^{-1}\mathfrak{p}$. AM 3.11.i gives us that $\mathfrak{b} = S^{-1}\mathfrak{a}$ is extended. Then $\mathfrak{p} \supset \mathfrak{a}$ implies $S^{-1}\mathfrak{p} \supset S^{-1}\mathfrak{a} = \mathfrak{b}$ whenever $f \notin \mathfrak{a}, \mathfrak{p}$, so that $V(\mathfrak{a}) \cap D(f) \subset \psi^{-1}(\mathfrak{b})$. Suppose that $S^{-1}\mathfrak{p} \supset S^{-1}\mathfrak{a}$. That is, for all $a \in \mathfrak{a}$, there is some $x \in \mathfrak{p}$ and $n \geqslant 1$ such that $f^n(a - x) = 0$. Hence, $f^n a \in \mathfrak{p}$, so that $\mathfrak{p}$ prime and $f \notin \mathfrak{p}$ gives us $a \in \mathfrak{p}$.We conclude that $\psi^{-1}(V(\mathfrak{b})) = V(\mathfrak{a}) \cap D(f) \subset D(f)$ is closed, so that $\psi = (\psi^*)^{-1}$ is continuous.

Note that $(\mathcal{O}_X|_{D(f)})_\mathfrak{p} \cong A_\mathfrak{p} \cong (A_\mathfrak{p})_f \cong (A_f)_{\psi(\mathfrak{p})} \cong (\mathcal{O}_Y)_{\psi(\mathfrak{p})}$ for each $\mathfrak{p} \not\ni f$. Call this isomorphism $\alpha_\mathfrak{p} : (\mathcal{O}_X|_{D(f)})_\mathfrak{p} \to (\mathcal{O}_Y)_{\psi(\mathfrak{p})}$, and let

$$\alpha : \sqcup_{\mathfrak{p}\in D(f)}(\mathcal{O}_X|_{D(f)})_\mathfrak{p} \to \sqcup_{\mathfrak{q}\in Y}(\mathcal{O}_Y)_\mathfrak{q}$$

be defined by $\alpha|_{(\mathcal{O}_X|_{D(f)})_\mathfrak{p}} = \alpha_\mathfrak{p}$. Then we have a map

$$\begin{aligned} \phi^\sharp(U) : \mathcal{O}_X|_{D(f)}(U) &\to (\phi^*)_*\mathcal{O}_Y(U) \\ s &\mapsto \alpha \circ s \circ \phi^*. \end{aligned}$$

We have that

$$\alpha \circ s|_V \circ \phi^* = \alpha \circ s \circ \phi^*|_{(\phi^*)^{-1}(V)}$$

as functions on $(\phi^*)^{-1}(V)$ whenever $V \subset U \subset D(f)$ are open and $s \in \mathcal{O}_X(U)$, so $\phi^\sharp(U)$ gives a sheaf homomorphism $\phi^\sharp : \mathcal{O}_X|_{D(f)} \to \mathcal{O}_Y$. We have that $\phi^\sharp_\mathfrak{p} = \alpha_\mathfrak{p}$ is an isomorphism of rings for all $\mathfrak{p} \in D(f)$, so $\phi^\sharp$ is itself a sheaf isomorphism. Then $(f, f^\sharp)$ is an isomorphism of locally ringed spaces, since ring isomorphisms preserve maximality.

□

2.2. Let $(X, \mathcal{O}_X)$ be a scheme, and let $U \subset X$ be any open subset. Show that $(U, \mathcal{O}_X|_U)$ is a scheme. We call this the *induced scheme structure* on the open set U, and we refer to $(U, \mathcal{O}_X|_U)$ as the open subscheme of U.

Proof. Choose a basic cover $\{D(f_i)\}$ of U. Then each

$$(D(f_i), (\mathcal{O}_X|_U)_{D(f_i)}) = (D(f_i), \mathcal{O}_X|_{D(f_i)}) \cong \operatorname{Spec} A_{f_i}$$

by 2.1. We conclude that $\{D(f_i)\}$ is an affine open cover of $(U, \mathcal{O}_X|_U)$, so that the latter locally ringed space is a sheaf. □

2.3. $(*)$ A scheme $(X, \mathcal{O}_X)$ is *reduced* if for every open set $U \subset X$, the ring $\mathcal{O}_X(U)$ has no nilpotent elements.

(a) Show that $(X, \mathcal{O}_X)$ is reduced if and only if for every $p \in X$, the local ring $\mathcal{O}_{X,p}$ as no nilpotent elements.

(b) Let $(X, \mathcal{O}_X)$ be a scheme. Let $(\mathcal{O}_X)_{\text{red}}$ be the sheaf associated to the presheaf $U \mapsto \mathcal{O}_X(U)_{\text{red}}$, where for any ring A, we denote by A_{red} the quotient of A by its nilradical $\mathfrak{N}(A)$. Show that $(X, (\mathcal{O}_X)_{\text{red}})$ is a scheme. We call it the *reduced scheme* associated to X, and denote it by X_{red}. Show that there is a morphism of schemes $X_{\text{red}} \to X$, which is a homeomorphism on the underlying topological spaces.

(c) Let $f : X \to Y$ be a morphism of schemes, and assume that X is reduced. Show that there is a unique morphism $g : X \to Y_{\text{red}}$ such that f is obtained by composing g with the natural map $Y_{\text{red}} \to Y$.

2.4. Let A be a ring, and let $(X, \mathcal{O}_X)$ be a scheme. Given a morphism $f : X \to \operatorname{Spec} A$, we have an associated map $f^\sharp : \mathcal{O}_{\operatorname{Spec} A} \to f_*\mathcal{O}_X$ on sheaves. Taking Global sections, we obtain a homomorphism $A \to \Gamma(X, \mathcal{O}_X)$. Thus, there is a natural map

$$\alpha : \operatorname{Hom}_{\mathfrak{Sch}}(X, \operatorname{Spec} A) \to \operatorname{Hom}_{\mathfrak{Ring}}(A, \Gamma(X, \mathcal{O}_X)).$$

Show that α is bijective.

Proof. We begin by showing the result for $X = \operatorname{Spec} B$ affine. In this case, α has an inverse map given by

$$\begin{aligned} \beta : \operatorname{Hom}_{\mathfrak{Ring}}(A, B) &\longrightarrow \operatorname{Hom}_{\mathfrak{Sch}}(\operatorname{Spec} B, \operatorname{Spec} A) \\ \varphi &\longmapsto (f, f^\sharp), \end{aligned}$$

where

$$\begin{aligned} f(\mathfrak{p}) &= \varphi^{-1}(\mathfrak{p}), \\ f^\sharp(s)(\mathfrak{p}) &= \varphi_{f(\mathfrak{p})}(s(f(\mathfrak{p}))). \end{aligned}$$

It suffices to show that $\beta = \alpha^{-1}$. Fix $(f, f^\sharp) \in \operatorname{Hom}_{\mathfrak{Sch}}(\operatorname{Spec} B, \operatorname{Spec} A)$, and denote

$$(\beta \circ \alpha)(f, f^\sharp) = (g, g^\sharp).$$

Let $\phi_{f(\mathfrak{p})} : A \to A_{f(\mathfrak{p})}$ and $\phi'_{\mathfrak{p}} : B \to B_{\mathfrak{p}}$ be the natural maps. The commutative square

$$\begin{tikzcd} A \arrow[r, "f^\sharp(\operatorname{Spec} A)"] \arrow[d, "\phi_{f(\mathfrak{p})}"'] & B \arrow[d, "\phi'_{\mathfrak{p}}"] \\ A_{f(\mathfrak{p})} \arrow[r, "f^\sharp_{f(\mathfrak{p})}"'] & B_{\mathfrak{p}} \end{tikzcd}$$

gives us that

$$\begin{aligned} g(\mathfrak{p}) &= (f^\sharp(\operatorname{Spec} A))^{-1}(\mathfrak{p}) \\ &= (\phi'_{\mathfrak{p}} \circ f^\sharp(X))^{-1}(m_{\mathfrak{p}}) \\ &= (f^\sharp_{f(\mathfrak{p})} \circ \phi_{f(\mathfrak{p})})^{-1}(m_{\mathfrak{p}}) \\ &= \phi^{-1}_{f(\mathfrak{p})}(m_{f(\mathfrak{p})}) \\ &= f(\mathfrak{p}), \end{aligned}$$

since $f^\sharp$ is a local homomorphism. Then

$$\begin{aligned} g^\sharp(s)(\mathfrak{p}) &= f^\sharp(X)_{g(\mathfrak{p})}(s(g(\mathfrak{p}))) \\ &= f^\sharp_{f(\mathfrak{p})}(s(f(\mathfrak{p}))) \\ &= f^\sharp(s)(\mathfrak{p}). \end{aligned}$$

implies $(g, g^\sharp) = (f, f^\sharp)$.

Fix $\varphi : A \to B$, and denote $\beta(\varphi) = (f, f^\sharp)$, $(\alpha \circ \beta)(\varphi) = \psi$. Recall that $a \in A$ is identified with $\mathfrak{p} \mapsto a_\mathfrak{p} \in A_\mathfrak{p}$, and similarly for $b \in B$. Thus, $\psi(a) = \varphi(a)$ iff they're equal as maps $\operatorname{Spec} B \to \sqcup_{\mathfrak{p} \in \operatorname{Spec} B} B_\mathfrak{p}$. The commutative square

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[d] & B \arrow[d] \\ A_{f(\mathfrak{p})} \arrow[r, "\varphi_{f(\mathfrak{p})}"'] & B_\mathfrak{p} \end{tikzcd}$$

gives us

$$\psi(a)(\mathfrak{p}) = \varphi_{f(\mathfrak{p})}(a(f(\mathfrak{p}))) = \varphi_{f(\mathfrak{p})}(a_{f(\mathfrak{p})}) = \varphi(a)(\mathfrak{p}).$$

Hence, $(\alpha \circ \beta)(\varphi) = \varphi$. We conclude that $\beta = \alpha^{-1}$, so α is a bijection.

Now, we consider the general case. Fix an affine cover $\{U_i \cong \operatorname{Spec} B_i\}$ of X. Suppose $f : X \to \operatorname{Spec} A$ is a morphism of schemes. Then it restricts to a set of morphisms $f_i : \operatorname{Spec} B_i \to \operatorname{Spec} A$ of affine schemes. If $f, g : X \to \operatorname{Spec} A$ are two morphisms of schemes with $\alpha(f) = \alpha(g)$, □

2.5. Describe $\operatorname{Spec} \mathbb{Z}$, and show that it is a final object for the category of schemes, i.e., each scheme X admits a unique morphism to $\operatorname{Spec} \mathbb{Z}$.

Proof. The only primes of $\mathbb{Z}$ are (p) for p prime, and (0). The former are all maximal, while the latter is a generic point for $\operatorname{Spec} \mathbb{Z}$, since $V(\mathfrak{a}) \ni (0)$ implies $(0) \supset \mathfrak{a}$, i.e. $\mathfrak{a} = (0) \subset \mathfrak{p}$ for all $\mathfrak{p}$ prime.

We have a bijection

$$\operatorname{Hom}_{\mathfrak{Sch}}(X, \operatorname{Spec} \mathbb{Z}) \to \operatorname{Hom}_{\mathfrak{Ring}}(\mathbb{Z}, \Gamma(X, \mathcal{O}_X)),$$

so that the former is singleton for all X by the fact that $\mathbb{Z}$ is initial in $\mathfrak{Ring}$. □

2.7. Let X be a scheme. For any point $x \in X$, let $\mathcal{O}_x$ be the local ring at x, and m_x its maximal ideal. We define the *residue field* of x on X to be the field $k(x) = \mathcal{O}_x/m_x$. Now, let K be any field. Show that to give a morphism of $\operatorname{Spec} K$ to X, it is equivalent to giving a point $x \in X$ and an inclusion map $k(x) \to K$.

Proof. Note that $\operatorname{Spec} K = \{(0)\}$ is singleton, so a morphism $f : \operatorname{Spec} K \to X$ is a point $x = f((0)) \in X$, as well as the information $f^\sharp(U) : \mathcal{O}_X(U) \to K$ for every $x \in U \subset X$ open. Restriction is the identity on $\mathcal{O}_{\operatorname{Spec} K}$, so this defines a map $f^\sharp_x : \mathcal{O}_x \to K$. Since each element of m_x is not invertible, this gives us a map $\overline{f^\sharp_x} : k(x) \to K$. It is a morphism of fields, and thus an injection.

Conversely, suppose we have an inclusion $k(x) \hookrightarrow K$. Then, for each $x \in U \subset X$, we have a map

$$f^\sharp(U) : \mathcal{O}(U) \to \mathcal{O}_x \to k(x) \hookrightarrow K.$$

These maps play well with restriction, thus giving us a morphism $f^\sharp : \mathcal{O}_X \to \mathcal{O}_{\operatorname{Spec} K}$ of sheaves. Defining $f : \operatorname{Spec} K \to X : (0) \mapsto x$ gives us a morphism $(f, f^\sharp) : \operatorname{Spec} K \to X$ of schemes, as desired. □

2.8. (∗) Let X be a scheme. For any point $x \in X$, we define the *Zariski tangent space* T_x at $x \in X$ to be the dual of the $k(x)$-vector space m_x/m_x^2. Now, assume that X is a scheme over a field k, and let $k[\epsilon]/\epsilon^2$ be the *ring of dual numbers* over k. Show that to give a k-morphism of $\operatorname{Spec} k[\epsilon]/\epsilon^2$ to X is equivalent to giving a point $x \in X$, *rational over* k (i.e. such that $k(x) = k$), and an element of T_x.

2.9. If X is a topological space, and Z an irreducible closed subset of X, a generic point for Z is a point ζ such that $Z = \{\zeta\}^-$. If X is a scheme, show that every (nonempty) irreducible closed subset has a unique generic point.

Proof. Let Z be an irreducible closed subset of a scheme X, and let $\{U_i \cong \operatorname{Spec} A_i\}_{i \in I}$ be an affine open cover of X. Then $Z = \cup_{i \in I} U_i \cap Z$ □

2.10. Describe $\operatorname{Spec} \mathbb{R}[x]$. How does its topological space compare to the set $\mathbb{R}$? To $\mathbb{C}$?

Proof. □

2.13. A topological space is *quasi-compact* if every open cover has a finite sub-cover.
  (a) Show that a topological space X is noetherian if and only if every open subset is quasi-compact.
  (b) If A is a noetherian ring, show that $^($ $\operatorname{Spec} A)$ is a noetherian topological space.
  (c) Give an example to show that $^($ $\operatorname{Spec} A)$ can be noetherian even when A is not.